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HAIR REMOVAL COMPOSITION DISPENSABLE FROM A SPRAY CAN

Abstract

Pulling-off hair removal composition dispensable from a spray can has at least about 45% by weight, relative to the total weight of the composition, of a wax; and from about 20% to about 45% by weight, relative to the total weight of the composition, of a propellant, which is gaseous at a temperature of 25° C. and at a pressure of 1 bar and liquid at a temperature of 25° C. and at a pressure of 2.8 bar. The propellant includes at least one organic compound chosen from the group consisting of C.sub.1-C.sub.4 aliphatic compounds, C.sub.2-C.sub.6 ethers and a combination thereof. The composition is easy to use and, at the same time, allows good hair removal to be obtained.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is related to Italian Patent Application No. 102022000007823 filed on Apr. 20, 2022, and to U.S. Provisional Application No. 63/456,715 filed on Apr. 3, 2023, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a hair removal composition and a use of such a composition. The present invention also relates to a hair removal product (containing the mentioned composition) and a kit comprising said product.

BACKGROUND OF THE INVENTION

[0003] In the field of hair removal it is known to use a wax that is heated so that it assumes a fluid consistency and can be applied, still hot, on an area of the skin to be treated. At this point, a layer of nonwoven fabric is made to adhere onto it with a little pressure. Once re-solidified, the wax is removed by pulling the nonwoven fabric swiftly in the opposite direction to the hair growth so that, while it is being pulled off, it removes the same hair (by grabbing it therewith).

[0004] This procedure, being relatively complex (mainly due to the need to heat the wax), is rarely used in the domestic environment and has some further drawbacks, including the fact that, when applied hot, it can create, to predisposed subjects, rashes on the skin.

[0005] In the domestic environment, therefore, so-called depilatory strips are more often used, i.e. support strips (made of paper, plastic and/or fabric) on the surface of which a layer of a cold active wax is arranged. Typically, these strips are marketed coupled i.e. like two support strips connected by an intermediate layer of wax. When it is wished to use the strips they are separated and the respective layer of glue remains on the surface of each one.

[0006] Although the depilatory strips are easier to use, they are also not free of drawbacks: it happens with a certain frequency that the wax dries (even only partially) and becomes ineffective; during the separation phase, it is not uncommon for one or both strips to break or for the resin to distribute unevenly between the two strips.

[0007] Aim of the present invention is to provide a hair removal composition, a use of such composition, a hair removal product and a kit comprising such a product, which allow to overcome, at least partially, the drawbacks of the prior art and are, at the same time, of easy and economical realization and/or use.

SUMMARY

[0008] According to the present invention there is provided a hair removal composition, a use of such a composition, a hair removal product and a kit comprising such a product as recited in the independent claims which follow and, preferably, in any one of the claims directly or indirectly dependent on the independent claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The invention is described below with reference to the accompanying figures, which show non-limiting embodiments thereof, wherein:

[0010] FIGS. **1** to **4** show an efficacy proof of an embodiment of the composition in accordance with the present invention relative to depilatory strips currently on the market;

[0011] FIG. **5** is a perspective view of a hair removal product in accordance with the present invention;

[0012] FIG. **6** is a section of a part of the product of FIG. **5**;

[0013] FIG. **7** is a section of a further part of the product of FIG. **5**;

[0014] FIGS. **8** to **10** show a test of efficacy of a comparison product; and

[0015] FIGS. **11** to **13** show a test of the efficacy of an embodiment of the composition in accordance with the present invention;

[0016] FIGS. **14** to **16** show a test of efficacy of an embodiment of the composition in accordance with the present invention; and

[0017] FIGS. **17** to **19** show a test of efficacy of an embodiment of the composition in accordance with the present invention.

DETAILED DESCRIPTION

[0018] In accordance with a first aspect of the present invention, there is provided a pulling-off composition. The composition, which, in particular, is dispensable from a spray can (see FIGS. **5-7** wherein the can, also referred to as a hair removal product, is identified by reference numeral **1**), comprises at least about 45% (in particular, at least about 50%; more in particular, at least about 55%; still more in particular, at least about 70%) by weight, relative to the total weight of the composition, of a wax; and from about 20% (in particular, from about 25%) to about 45% (in particular, to about 40%) by weight, relative to the total weight of the composition, of a propellant, which is gaseous at a temperature of 25° C. and at a pressure of 1 bar and liquid at a temperature of 25° C. and at a pressure of 2.8 bar. The propellant comprises (in particular, consists of) at least one organic compound chosen from the group consisting of: C.sub.1-C.sub.4 aliphatic compounds (in particular, alkyls), C.sub.2-C.sub.6 ethers and a combination thereof (i.e. mixture).

[0019] It has been experimentally observed that a composition of this type allows to obtain a hair removal of comparable quality, even better than the known techniques. At the same time, the mentioned composition is easy to use and reduces the risk of waste.

[0020] In this regard, it has been noted that the propellant, addition to allowing a correct dispensing of the in composition (from the spray can **1**) keeps the wax liquid during application (thus favouring the usability thereof) and, by evaporating (at least partially), allows the wax to (re) solidify.

[0021] Unless explicitly stated otherwise, in the present text, “C.sub.x-C.sub.y” refers to a group and/or a compound which is understood as having x to y carbon atoms.

[0022] Unless explicitly stated otherwise, “aliphatic” in this text means a non-aromatic and unsubstituted (unless otherwise specified), saturated or unsaturated, linear, branched and/or cyclic hydrocarbon. Non-limiting examples of aliphatic groups are: t-butyl, ethenyl, 1- or 2-propenyl, cyclohexyl.

[0023] Unless explicitly stated otherwise, by “alkyl” or “alkyl-” it is meant a saturated aliphatic (i.e. an aliphatic group without double or triple carbon-carbon bonds). Non-limiting examples of alkyls are: methyl, n-propyl, t-butyl, cyclohexyl.

[0024] By pulling-off hair removal is meant the type of hair removal that involves making a product to adhere to an area of the body (so that the product sticks to the hair) and then removing it by swiftly pulling it so that, while it is being pulled off, it removes the hair by grabbing therewith.

[0025] Advantageously but not necessarily, by wax is meant a substance substantially solid at a temperature lower than about 40° C. (in particular, lower than about 60° C.; more in particular

lower than about 65° C.) at a pressure of 1 bar. In addition or alternatively, the wax is a substance (the substance is) substantially liquid at the temperature of about 100° C. (in particular, at the temperature of about 90° C.) (in particular, at a 1 bar pressure).

[0026] According to some non-limiting embodiments, the wax has a softening point (at 1 bar) greater than about 40° C. (in particular, greater than about 70° C.; in particular, at about 80° C.). Alternatively or additionally, the wax has a softening point (at 1 bar) lower than about 110° C. (in particular, lower than about 100° C.).

[0027] According to some non-limiting embodiments, the wax comprises (in particular, is) a resin (in particular, substantially liquid at the temperature of about 100° C.—at a 1 bar pressure). In some specific non-limiting cases, the resin has a softening point (at 1 bar) ranging from about 70° C. (in particular, about 80° C.; more particularly, about 85° C.) to about 99° C. (in particular, to about 95° C.; more in particular, to about 90° C.).

[0028] The following procedure was followed to measure the softening point of a material (e.g. a wax and/or a resin). A standard-sized ring is hot-filled with resin, which is then cooled. The full ring is placed on a support inside a thermostatic oil bath. Above the full ring there is placed a steel ball of standardized size. Heating begins. The resin slowly softens and no longer resists the weight of the ball that pushes the resin downwards. When the ball touches the base of the support, the temperature is read.

[0029] More precisely, to measure the softening point of a material (e.g. a wax and/or a resin), the provisions of ASTM E28 regulation (July 2018 edition) were followed.

[0030] Typically, moreover, the wax is substantially insoluble in water and soluble in an organic solvent.

[0031] Advantageously but not necessarily, the propellant is gaseous also at a temperature of about 0° C. (in particular, also at a temperature of about -20° C.) and at a pressure of 1 bar. Alternatively or additionally, the propellant is substantially liquid at a temperature of about 25° C. and also at a pressure of about 2.5 bar (in particular, also at a pressure of about 2 bar).

[0032] According to some non-limiting embodiments, the propellant has a boiling point between about -45° C. (in particular, about -30° C.; more in particular, about -25° C.) and 0° C. (in particular, about -10° C.).

[0033] To measure the boiling point of a substance (such as of the propellant), it is proceeded in this way. In a test tube, one pours 15-20 ml of the liquid substance to be analysed and a glass capillary (with an inner diameter of about 1 mm and a wall thickness of about 0.1-0.2 mm) open at an overturned end (i.e. with the closed part turned upwards, and the open part drawing in the substance). At this point, an air bubble occupying most of the capillary so positioned inside a test tube is observed. With a clamp, the thermometer is positioned at the same height as the substance contained in the tube and the whole is immersed in a beaker containing a further liquid. It is made sure that the level of the liquid is higher than the level of the substance to be analysed. It is heated and during heating bubbles escaping from the mouth of the overturned capillary can be observed. As soon as the rapid and continuous escape of bubbles (two or more bubbles per second) can be observed, the heating is interrupted. It is allowed to cool slowly and the mouth of the capillary is observed. The boiling point detected is the temperature at which the flow of bubbles slows down, until it disappears, and the substance starts rising inside the capillary.

[0034] More precisely, to measure the boiling point of a substance (for example of the propellant) it is proceed as required by ASTM D86-20b standard (2020 edition).

[0035] Advantageously but not necessarily, the composition comprises up to about 80% (in particular, up to about 75%; more in particular, up to about 70%; still more in particular, up to about 65%) by weight, relative to the overall weight of the composition, of the wax.

[0036] According to specific non-limiting embodiments, the composition comprises from about 50% to about 70% (in particular, to about 65%; more in particular, to about 60%) by weight, relative to the overall weight of the composition, of the wax.

[0037] Advantageously but not necessarily, the propellant comprises at least about 10% by weight (in particular, at least about 20%; more in particular, at least about 35%), relative to the weight of the propellant, of at least one ether chosen among C.sub.2-C.sub.6 ethers. In other words, the aforementioned organic compound is chosen among C.sub.2-C.sub.6 ethers.

[0038] Advantageously but not necessarily, the organic compound is chosen among C.sub.2-C.sub.4 ethers (in particular, the dimethyl ether).

[0039] According to some embodiments, the propellant comprises up to about 60% by weight (in particular, up to about 50%; more in particular, about 45%), relative to the weight of the propellant, of the aforementioned organic compound chosen among C.sub.2-C.sub.6 ethers (in particular among C.sub.2-C.sub.4 ethers; more in particular, the dimethyl ether).

[0040] In particular, in these cases, the propellant may or may not also comprise one or more aliphatic compounds chosen among the C.sub.1-C.sub.4 aliphatic compounds.

[0041] It has been experimentally observed that the use of the ether (in particular, of the dimethyl ether) allows to obtain a better (and much faster) solubilization of wax.

[0042] Advantageously but not necessarily, the propellant comprises at least about 15% (in particular, at least about 40%; more in particular, at least about 50%) by weight, relative to the weight of the propellant, of one or more of the mentioned C.sub.1-C.sub.4 aliphatic compounds. In particular, in these cases, the propellant may or may not also comprise one or more ethers chosen among C.sub.2-C.sub.6 ethers (in particular, C.sub.2-C.sub.4; more in particular, dimethyl ether).

[0043] According non-limiting embodiments, the C.sub.1-C.sub.4 aliphatic compounds are C.sub.1-C.sub.4 alkyls (in particular, the C.sub.1-C.sub.4 aliphatic compounds are chosen among propane, butane and a combination thereof).

[0044] In particular, in the present text by “combination” of different compounds is meant a mixture thereof.

[0045] Advantageously but not necessarily, the propellant comprises up to about 90% by weight (in particular, up to about 70%; more in particular, up to about 65%), relative to the weight of the propellant, of one or more of the C.sub.1-C.sub.4 aliphatic compounds (in particular, C.sub.1-C.sub.4 alkyls; more in particular, propane and/or butane).

[0046] According to some non-limiting embodiments, the propellant comprises from about 15% (in particular, from about 40%; more in particular, from about 50%) to about 90% (in particular, to about 70%; more in particular, to about 65%), relative to the weight of the propellant, of one or more of the C.sub.1-C.sub.4 aliphatic compounds (in particular, C.sub.1-C.sub.4 alkyls; more in particular, propane and/or butane); and from about 10% by weight (in particular, from about 20%; more in particular, from about 35%) to about 60% by weight (in particular, to about 50%; more in particular, to about 45%), relative to the weight of the propellant, of at least one ether chosen among the C.sub.2-C.sub.6 ethers (in particular, among the C.sub.2-C.sub.4 ethers; and more in particular, the dimethyl ether).

[0047] Advantageously but not necessarily, the propellant is (in particular, predominantly-more particularly, for at least 90% by weight, relative to the total weight of the propellant itself) consisting of one or more of the C.sub.1-C.sub.4 aliphatic compounds (in particular, C.sub.1-C.sub.4 alkyls; more in particular, propane and/or butane) and of at least one ether chosen among the C.sub.2-C.sub.6 ethers (in particular among the C.sub.2-C.sub.4 ethers; more in particular, the dimethyl ether).

[0048] According to some examples, the propellant comprises propane and/or butane.

[0049] According to specific but non-limiting embodiments, the propellant comprises (in particular, consists of) from about 5% (in particular, from about 20%; more in particular, from about 35%; still more in particular, from about 40%) to about 55% (in particular, to about 50%) by weight, relative to the weight of the propellant, of butane (in particular, n-butane); and/or from about 10% to about 60% (in particular, to about 40%; more in particular, to about 20%) by weight, relative to the weight of the propellant, of propane; and from about 10% (in particular, from about 25%; more in

particular, from about 35%) to about 60% (in particular, to about 50%; more in particular, to about 45%) by weight, relative to the weight of the propellant, of dimethyl ether.

[0050] Advantageously but not necessarily, the composition consists (predominantly; in particular, for at least 70% by weight; more in particular, for at least 85% by weight, relative to the weight of the composition itself) of the aforementioned wax and of the aforementioned propellant.

[0051] According to some non-limiting embodiments, the aforementioned wax is a resin chosen from the group consisting of: Rosin (CAS No. 8050-09-7), glycerol ester of rosin (CAS No. 8050-31-5), glycerol ester of rosin stabilized through hydrogenation (CAS No. 65997-13-9), triethylene glycol ester of rosin (2,2'-[1,2-ethanediylbis(oxy)]bisethanol ester of rosin acids); synthetic polycyclopentadiene and hydrogenated polycyclopentadiene resins (CAS No. 68132-00-3/25568-84-7), synthetic hydrogenated styrene resins, methyl styrene resins, indene copolymers (1H-Indene, polymer with ethenylbenzene and (1-methyl ethenyl)benzene, hydrogenated) (and a combination thereof).

[0052] Advantageously but not necessarily, wax is a resin consisting predominantly (in particular, for at least about 80% by weight—more in particular, at least about 85% by weight, relative to the total weight of the resin) of abietic acid (CAS No.: 514-10-3) and/or esters thereof (in particular with glycerol) (possibly stabilised by hydrogenation).

[0053] According to some non-limiting embodiments, the resin further comprises (in particular, up to about 20%—more in particular, up to about 15%; still more in particular, up to about 10%—by weight, relative to the total weight of the resin, of) other resin acids and/or esters thereof (in particular, with glycerol).

[0054] In some cases, the resin comprises, at least 5% by weight, relative to the total weight of the resin, other resin acids and/or esters thereof (in particular, with glycerol).

[0055] In particular, the resin acids are chosen from the group consisting of: dihydroabietic acid (C.sub.20H.sub.32O.sub.2—CAS No. 19402-28-9), dehydroabietic acid (C.sub.20H.sub.28O.sub.2—CAS No. 1740-19-8), isopimaric acid (CAS No. 5835-26-7), pimaric acid (CAS No. 127-27-5) and a combination thereof.

[0056] According to specific non-limiting implementation forms, resin is chosen in the consistent group of: Rosin (CAS No. 8050-09-7), rosin glycerol ester (CAS No. 8050-31-5), resin acids, polycyclopentadiene (synthetic) resins (and a combination thereof). Specifically, resin includes (more specifically, it is) Rosin (CAS No. 8050-09-7), rosin glyceric ester (CAS No. 8050-31-5) polycyclopentadiene (synthetic) resins (and a combination thereof).

[0057] Advantageously but not necessarily, the composition comprises (in particular, from about 10% to about 15% by weight, relative to the total weight of the composition, of) an oil component (in particular, an oil or mixture of oils) chosen from the group consisting of: mineral oils (in particular white), vegetable oils (in particular, colza oil, corn oil, sunflower oil or a combination thereof) and C.sub.17-C.sub.39 esters (in particular, isopropyl myristate—CAS No. 110-27-0—and/or medium chain triglycerides—with C.sub.18-C.sub.36 fatty acids).

[0058] According to specific embodiments, the oil component comprises (in particular consists of) mineral oils (e.g. CAS No. 8042-47-5).

[0059] Advantageously but not necessarily, the composition comprises (in particular, from about 0.1% to about 5% by weight, relative to the total weight of the composition, of) at least one plasticizer.

[0060] According to some non-limiting embodiments, the plasticizer comprises (is) an ethylene-vinyl acetate copolymer (e.g., CAS No. 108-05-4).

[0061] Advantageously but not necessarily, the plasticizer has a melting point ranging from about 80° to about 110° C. (at 1 bar). Additionally or alternatively, the plasticizer has a density ranging from about 0.93 g/cm.sup.3 to about 0.97 g/cm.sup.3 (Method: ISO 1183:2019).

[0062] In accordance with a second aspect of the present invention, there is provided a hair removal product **1** (FIGS. 5-7). The product **1** is a container **2**, which, in turn, is provided with a

fluid-tight chamber **3** (FIG. **6**); a composition as described above in accordance with the first aspect of the present invention, which composition is arranged (in particular, under pressure) inside the fluid-tight chamber **3**; and a dispensing device **4** (FIGS. **5** and **7**), which is configured to dispense the composition present in the fluid-tight chamber towards the outside.

[0063] Advantageously but not necessarily, the dispensing device **4** is configured to turn the composition into foam while transferring the composition from the inside to the outside of the fluid-tight chamber **3**.

[0064] Advantageously but not necessarily, the pressure inside the fluid-tight chamber **3** is between 2.5 (in particular, 2.8) and 3.5 (in particular, 3.3) bar.

[0065] In accordance with a third aspect of the present invention, there is provided a hair removal kit comprising a product **1** as described above relatively to the second aspect of the present invention and at least one sheet (in particular, a strip) of flexible material.

[0066] According to some non-limiting embodiments, the flexible material is chosen among: paper, fabric, plastic, nonwoven fabric and a combination thereof.

[0067] Advantageously but not necessarily, the kit also comprises a spatula. In particular, the spatula is configured to lay (spread) the aforementioned composition (once applied) on an area of the body.

[0068] In accordance with a fourth aspect of the present invention, there is provided a use of the composition as described above relatively to the first aspect of the present invention for the removal of hair of the human body.

[0069] Further characteristics of the present invention will become apparent from the following description of some merely illustrative and non-limiting examples.

Example 1

[0070] This example describes the procedure for preparing a composition in accordance with the present invention.

[0071] The components used are shown in Table 1 below.

TABLE-US-00001	TABLE 1	Quantity (g)	DESCRIPTION	% Product	DERCOL GL 85	53.878										
323.277	(glycerol and rosin esters)	PHARMALUB 45	11	66.00	TITANIUM DIOXIDE	0.33 1.98										
ELVAX 250 A	0.528	3.17	SHEA BUTTER	0.066	0.40	HEMP OIL	0.066 0.40									
SCENT	0.132	0.79	COLOUR CI 26100	SOL 10%	0.066	0.40	IN OIL	PHARMALUB 45	Butane GAS	45%	- Propane	34	204.00	15%	- DME	40%

[0072] DERCOL GL 85 is a wax (in particular, a resin of resin acids and rosin acids, esters with glycerol—CAS No. 8050-31-5) whose supplier is Diamantino Malho & C, Lda; PHARMALUB 45 is a white mineral oil (CAS No. 8042-47-5) whose supplier is BT Commerciale S.r.l; the supplier of titanium oxide is UNIVAR SpA; ELVAX 250 A is a plasticizer (film-forming polymer) whose supplier is Cometechn (manufacturer: Dupont Performance Materials International Sàrl, 2 Chemin Du Pavillon, 1218 LE GRAND-SACONNEX, SWITZERLAND) and consists of an ethylene-vinyl acetate copolymer (N. CAS108-05-4); the supplier of SHEA BUTTER and HEMP OIL is ACEF SpA; the supplier of the SCENT is GRC Parfum SpA; the supplier of the propellant gas (combination of Butane, propane and DME—dimethyl ether) is SETTALA GAS Srl.

[0073] The following procedure was followed. In a Mixer the following components have been loaded in succession: DERCOL GL 85 (Resin), PHARMA 45 OIL (Pharmaceutical White Mineral Oil) and TITANIUM DIOXIDE.

[0074] It was heated to $120\pm 10^{\circ}$ C. under stirring until the semi-finished product had completely melted.

[0075] Once the melting and $120\pm 10^{\circ}$ C. were reached, the ELVAX 250A component was added and stirred until complete melting. The product was naturally cooled to $80\pm 3^{\circ}$ C.

[0076] Once the temperature of 80° C. was reached, the remaining components (Excipients, Colour, Scent) were added to the mixture, except for the propellant gas.

[0077] The aluminium cans were filled with a cosmetic doser heated to $80\pm 3^{\circ}$ C. and the shut-

off/dispensing valves were crimped.

[0078] At this point, the propellant gas was added and the cans were capped with the appropriate dispensers.

[0079] After five days of storage it was possible to check whether the resin had adequately solubilized by shaking the cans. When the balls could be heard moving inside the can, the product was considered ready for use.

Example 2

[0080] The functionality of the composition obtained as indicated in the previous example was tested with respect to Veet Easy-Gel™ depilatory strips.

[0081] The efficacy test consisted of the epilation of two delimited areas of skin. As can be inferred from the photographs (FIGS. 1-4) the hair removal takes place in an excellent and substantially identical way in both cases. In this context, it should be noted that: FIG. 1 shows the areas before hair removal; FIG. 2 shows a Veet depilatory strip applied on the left leg and the composition obtained according to the previous example applied on the right leg; FIG. 3 shows the left leg after the depilatory strip has been pulled off; and FIG. 4 shows the right leg after the mentioned composition has been pulled off.

[0082] From the tests carried out, the following was observed.

[0083] The cold strips must be mechanically heated before being peeled apart, this action is drastically affected by external temperature conditions. Poor heating of the strip causes the cold strips to be peeled apart incorrectly (the wax remains only on one side of the strip and is no longer effective). Excessive heating causes excessive fluidity of the wax that remains attached to the skin without removing the hair.

[0084] By repeating the tests several times, it was therefore noted that the results obtained with the composition of the previous example are more constant. The composition has no preparatory condition and is not affected by external temperature.

Example 3

[0085] This example describes the procedure for preparing a comparison product.

[0086] The components used are shown in Table 2 below.

TABLE-US-00002	TABLE 2	Quantity (g)	DESCRIPTION	% Product
	POLYSTYRENE RESIN	8.8	52.80	
	PINE RESIN	4.4	26.40	
	DIMETHYL ETHER (DME)	43	258.00	
	LPG	43	258.00	
	KAOLIN	0.6	3.60	
	PERFUME	0.2	1.20	
	TOTAL	100	600	

[0087] The components were sequentially loaded into a mixer: POLYSTYRENE RESIN, PINE RESIN, KAOLIN. It was heated to about 120° C. under stirring until the semi-finished product was completely melted. The product was cooled naturally, and when the temperature of 80° C. was reached, perfume was added.

[0088] After that, an aluminum cylinder was filled with the semi-finished product to which Propellant Gas (DME+LPG) was added. Once the closure valve was crimped, the cylinder was closed with the appropriate nozzle.

[0089] After a few days of storage, the ball contained inside the cylinder began to move, emitting a metallic sound that defines the product as ready for use.

Example 4

[0090] This example describes the comparison between the product obtained according to the previous example (comparison product) and the composition of Example 1.

[0091] The Evaluation Criteria follow a Reference Scale from 0 to 4, in ascending order of adequate compliance: [0092] 0=Not applicable (NA) [0093] 1=Not Sufficient [0094] 2=Sufficient [0095] 3=Good [0096] 4=Excellent

[0097] The tests performed consisted of the application and on the skin of an area of an arm, overlapping of a depilatory strip and subsequent tearing for epilation. The results obtained are shown in Table 3 below.

TABLE-US-00003	TABLE 3	ASSESSMENT	Product of Composition of	CRITERION	example 3
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example 1 Dispensing 1 4 Drafting 0 4 Adhesion 0 4 Tearing 0 4 Effectiveness 0 4

[0098] The pictures in FIGS. **8-10** are photographs of the tests with the (comparison) product of Example 3 while FIGS. **11-13** are photographs showing different stages of the tests with the composition of Example 1.

Example 5

[0099] This example describes the preparation of additional compositions in accordance with the present invention.

[0100] The components used are shown in the following Tables 4-7. The procedure followed for the preparations is the same as described in Example 1 above.

TABLE-US-00004 TABLE 4 (composition H313/1) Quantity (g) DESCRIPTION % of Product
DERCOL GL-85 67.878 407.27 PHARMALUB 45 11 66.00 TITANIUM DIOXIDE 0.33 1.98
ELVAX 250 A 0.528 3.17 SHEA BUTTER/CHLOROPHYLL 0.066 0.40 PERFUME 0.132 0.79
COLOR 0.066 0.40 PROPELLENT GAS 20 120.00 TOTAL 100 600

TABLE-US-00005 TABLE 5 (composition H314/1) Quantity (g) DESCRIPTION % of Product
DERCOL GL-85 45.878 275.27 PHARMALUB 45 8 48.00 TITANIUM DIOXIDE 0.33 1.98
ELVAX 250 A 0.528 3.17 SHEA BUTTER/CHLOROPHYLL 0.066 0.40 PERFUME 0.132 0.79
COLOR 0.066 0.40 PROPELLENT GAS 45 270.00 TOTAL 100 600

TABLE-US-00006 TABLE 6 (composition H315/1) Quantity (g) DESCRIPTION % of Product
COLOPHONY 53.878 323.27 PHARMALUB 45 11 66.00 TITANIUM DIOXIDE 0.33 1.98
ELVAX 250 A 0.528 3.17 SHEA BUTTER/CHLOROPHYLL 0.066 0.40 PERFUME 0.132 0.79
COLOR 0.066 0.40 PROPELLANT GAS 34 204.00 TOTAL 100 600

TABLE-US-00007 TABLE 7 (composition H316/1) Quantity (g) DESCRIPTION % of Product
POLYCYCLOPENTADIENE 53.878 323.277 PHARMALUB 45 11 66.00 TITANIUM DIOXIDE
0.33 1.98 ELVAX 250 A 0.528 3.17 SHEA BUTTER/CHLOROPHYLL 0.066 0.40 PERFUME
0.132 0.79 COLOR 0.066 0.40 PROPELLANT GAS 34 204.00 TOTAL 100 600

[0101] The propellant gas is: GAS Butane 45%-Propane 15%-DME 40%.

Example 6

[0102] This example describes tests performed on the compositions in Example 5.

[0103] The evaluation criteria and procedures followed are as those given in Example 4.

[0104] The results obtained for the H313/1 and H314/1 compositions are summarized in Table 8.

[0105] The pictures in FIGS. **14-16** are photographs of the tests with the H313/1 composition while FIGS. **17-19** are photographs showing different stages of the tests with the H314/1 composition.

TABLE-US-00008 TABLE 8 ASSESSMENT Composition Composition CRITERION H313/1
H314/1 Dispensing 3 4 Spreading 3 4 Adhesion 4 4 Tearing 4 3 Effectiveness 4 3

[0106] Composition H313/1 is characterized by adequate delivery of the semi-finished product. The wax appears to have a more compact, apparently “harder” foaminess than the original formulation (Propellant Gas percentage of 34% by weight). This consistency, having an excellent and adequate viscosity, ensures good and even spreading. The wax adheres properly to the hair and not to the skin, allowing effective hair removal with tearing that turns out to be more firm. The result is flawless.

[0107] Composition H314/1 presents a very good delivery of the semi-finished product. The foaminess is less compact, more “liquid-like” (due to the higher percentage of gas unlike the H313/1 test) than the original formulation (Propellant Gas percentage of 34% by weight) but equally good in the acceptable ranges for excellent application. Perfect and even spreading allows good adhesion of the wax to the hair, ensuring an effective result. The soft and gentle tear allows total removal of the hair.

[0108] The results obtained for the H315/1 and H316/1 compositions are summarized in Table 9.

TABLE-US-00009 TABLE 9 ASSESSMENT CRITERION H315/1 H316/1 Dispensing 4 4
Spreading 4 4 Adhesion 4 4 Tearing 4 4 Effectiveness 4 4

[0109] In composition H315/1, the resin used (rosin) is a natural resin, derived from the tear of the

Pine tree (Gem) and subsequently purified by distillation. It has a strong and pungent characteristic odor and a fluid and very soft consistency. In the formulation, Rosin, makes effective tearing with the formation of an elastic film upon removal of the hair. With the presence of rosin in the formula, the wax does not dry out but remains soft and elastic, allowing for better spreadability and easy cold use. Its special elasticity minimizes the presence of wax residue on the skin.

[0110] The H316/1 composition with Polycyclopentadiene allows stronger and more effective tearing. The special formulation based on synthetic resin, which is highly purified and performs well in tearing, constitutes a product that respects the epidermis, giving firm and hypoallergenic hair removal. The use of the composition with Polycyclopentadiene ensures accurate and extremely effective hair removal.

[0111] The results obtained for the H315/1 and H316/1 compositions are similar to those obtained for the composition in Example 1.

[0112] Note that compositions with different types of propellants were also tested, such as: mixtures of butane and DME (55% and, respectively, 45% by weight); mixtures of propane and DME (40% and, respectively, 60% by weight); and mixtures of butane and propane (65% and, respectively, 35% by weight). In these cases, the results obtained were also good (compositions with mixtures of butane and propane only took longer to solubilize the wax).

Claims

1. A pulling-off hair removal composition comprising: at least about 45% by weight, relative to the total weight of the composition, of a wax; and from about 20% to about 45% by weight, relative to the total weight of the composition, of a propellant, which is gaseous at a temperature of 25° C. and at a pressure of 1 bar and liquid at a temperature of 25° C. and at a pressure of 2.8 bar; wherein the propellant comprises at least one organic compound chosen from the group consisting of C.sub.1-C.sub.4 aliphatic compounds, C.sub.2-C.sub.6 ethers, and a combination thereof; wherein the composition is dispensable from a spray can.
2. The composition according to claim 1, wherein the wax is a resin that is solid at a temperature below about 40° C. at a pressure of 1 bar.
3. The composition according to claim 1, wherein the wax is present up to about 80% by weight, relative to the total weight of the composition, wherein the wax is substantially solid at a temperature lower than about 40° C. and at a pressure of 1 bar; and the propellant is chosen among the C.sub.2-C.sub.4 ethers.
4. The composition according to claim 3, wherein said organic compound is dimethyl ether.
5. The composition according to claim 1, wherein the wax is substantially solid at a temperature lower than about 65° C. and at a pressure of 1 bar and comprises a resin; and the propellant is gaseous at a temperature of about 0° C. and at a pressure of about 1 bar.
6. The composition according to claim 1, wherein the propellant comprises at least about 15% by weight, relative to the weight of the propellant, of said C.sub.1-C.sub.4 aliphatic compounds; said C.sub.1-C.sub.4 aliphatic compounds are C.sub.1-C.sub.4 alkyls.
7. The composition according to claim 1, wherein the propellant comprises from about 5% to about 55% by weight, relative to the weight of the propellant, of butane; from about 10% to about 60% by weight, relative to the weight of the propellant, of propane; and from about 10% to about 60% by weight, relative to the weight of the propellant, of dimethyl ether.
8. The composition according to claim 1, wherein said wax is a resin that is substantially liquid at a temperature of 100° C.
9. The composition according to claim 1, wherein said wax is a resin chosen from the group consisting of: rosin, glycerol ester of rosin, glycerol ester of rosin stabilized through hydrogenation, triethylene glycol ester of rosin, (2,2'-[1,2-ethanediylbis(oxy)]bisethanol ester of rosin acids); synthetic polycyclopentadiene and hydrogenated polycyclopentadiene resins, synthetic

hydrogenated styrene resins, methyl styrene resins, indene copolymers, and combinations thereof.

10. The composition according to claim 1, wherein said wax is a resin consisting of abietic acid and/or esters of abietic acid.

11. The composition according to claim 1, further comprising an oil component chosen from the group consisting of: mineral oils, plant oils and C.sub.17-C.sub.39 esters.

12. The composition according to claim 1, further comprising ethylene-vinyl acetate copolymer.

13. A hair removal product comprising: a container defining a fluid-tight chamber; a composition according to claim 1, inside said fluid-tight chamber; and a dispensing device configured to dispense said composition present in said fluid-tight chamber towards the outside.

14. The product according to claim 13, wherein the dispensing device is configured to turn said composition into foam while transferring the composition from the inside to the outside of said fluid-tight chamber; and wherein the pressure inside said fluid-tight chamber ranges from 2.5 to 3.5 bar.

15. A hair removal kit comprising a product according to claim 13 and at least one sheet of flexible material.

16. (canceled)

17. The composition according to claim 3, wherein the C.sub.2-C.sub.4 ethers is present in a range of about 20% by weight to about 25% by weight of the propellant.

18. The composition according to claim 6, wherein the C.sub.1-C.sub.4 aliphatic compounds are selected from the group consisting of propane, butane and a combination thereof.

19. The composition according to claim 8, wherein said has a melting point ranging from about 70° C. to about 99° C.

20. The composition according to claim 11, wherein the mineral oils comprise white oil, the plant oils is selected from the group consisting of colza oil, corn oil, sunflower oil, and combinations thereof, and the C.sub.17-C.sub.39 esters comprise isopropyl myristate and/or medium-chain triglycerides.
